EE 2001 Digital Systems Quiz I

Monday, 19/02/2024

Closed book, 50 mins, 20 points

Remember…

- *Make reasonable assumptions when in doubt. However, you should show all steps and justify assumptions for full credit.*
- *The final answers should be written in the space provided; otherwise it will not be evaluated*

1. (a) Given two binary strings A =10011101 and B =11101100, evaluate the 8-bit result after NOR and XOR logical operations respectively. (2.5 pts)

NOR: 00000010 XOR: 01110001

(b) Perform the binary equivalent of signed decimals (+48) + (-34) and (-34) + (-48) using 2's-complement with just enough digits to represent them. (2.5 pts)

+48 = 0110000

-48 = 1001111 +1 = 1010000

+34 = 0100010

-34 = 1011101 +1 = 1011110

+48 + (-34)

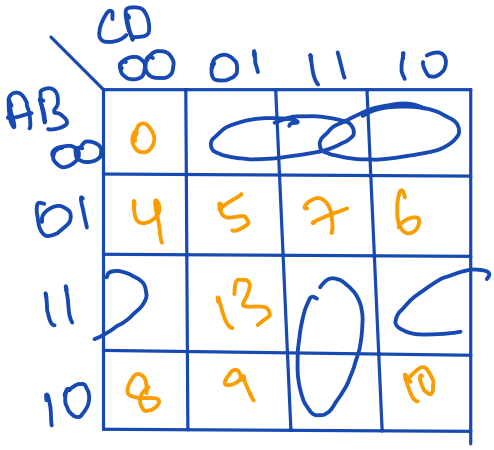
```
  1 1
  0110000
+ 1011110
----------
  1000110   (+14)
```

(-34) + (-48)

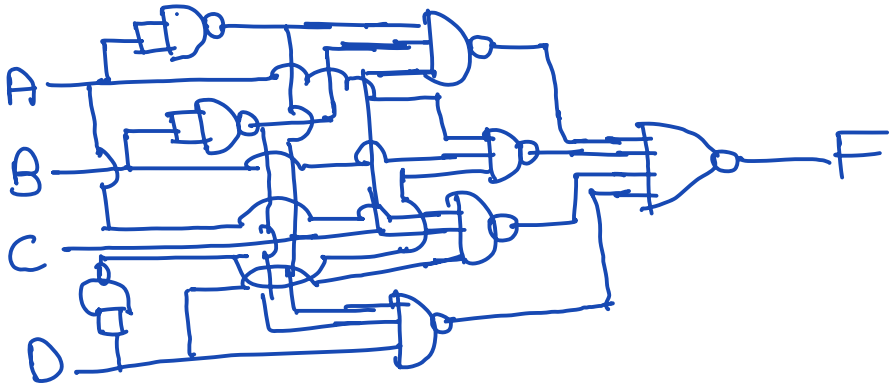
```
  1 1
  1011110
+ 1010000
----------
  1010110   (-82)
```

2. Determine all the essential prime implicants for the following Boolean function, and realize the simplified expression using two-level NAND circuit. Assume that complementary inputs as well as multi-input gates are available. (5 pts)

F (A,B,C,D) = Π (0, 4, 5, 6, 7, 8, 9, 10, 13)



$A'B'C + A'B'D + ACD + ABD'$



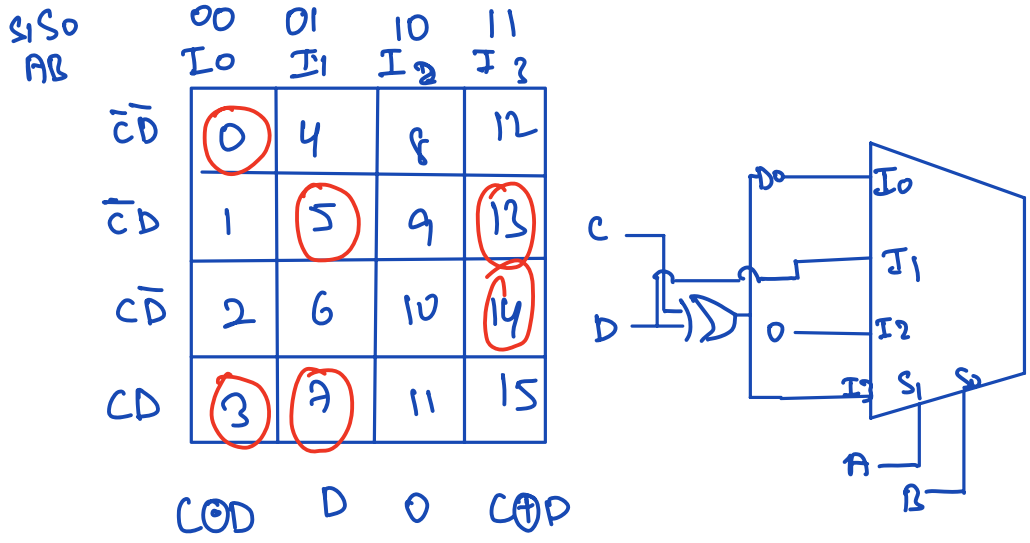
3. With the help of a truth table, design a combinational circuit that converts a four-bit binary number to a four-bit Gray code, and implement it with exclusive-OR gates.

(5 pts)

4. Implement the following Boolean function with the help of a multiplexer.

F (A, B, C, D) = Σ(0, 3, 5, 7, 13, 14)

(5 pts)



1. In a particular number system of base r, the roots of the equation $5x^2 + 50x + 125 = 0$ are given by $x = -8$ and $x = -5$. The value of r is 13 (2 pts)

$(x+8)(x+5)=0$

$x^2+13x+40=0$

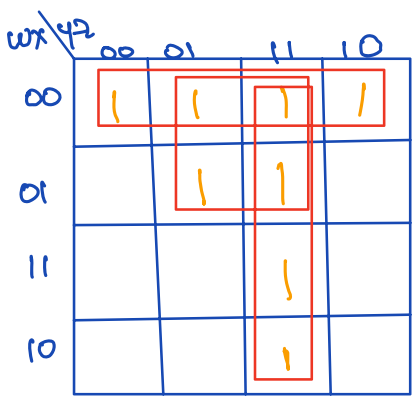
$5x^2+65x+200=0$

$5r=65$

$r^2+2r+5=200$ $\}\ r=13$

2. Simplify the following Boolean function and implement the same using NAND-AND gates:

$F(w, x, y, z) = \Pi(4, 6, 8, 9, 10, 12, 13, 14)$ (3 pts)



$wx' + yz + w'z$